Homework 6

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Section 003
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Collaboration Statement

Problem 1 *(Ross Chapter 2, 9.3)*

Suppose $\lim a_n = a$, $\lim b_n = b$, *and* $s_n = \dfrac{a_n^3 + 4a_n}{b_n^2 + 1}$. *Prove that* $\lim s_n = \dfrac{a^3 + 4a}{b^2 + 1}$ *carefully, using the limit theorems.*

Discussion. Here we essentially want to break the problem up into the individual limit components and apply the limit theorems 9.1-9.7 accordingly to show that the limit converges to $\frac{a^3+4a}{b^2+1}$.

For reference, here is what each of the 9.1-9.7 theorems say in a nutshell:

9.1 Theorem: Convergent sequences are bounded.

9.2 Theorem: If $\lim s_n = s$, then $\lim(ks_n) = k \lim s_n$.

9.3 Theorem: If $\lim s_n = s$ and $\lim t_n = t$, then $\lim(s_n + t_n) = \lim s_n + \lim t_n$.

9.4 Theorem: If $\lim s_n = s$ and $\lim t_n = t$, then $\lim(s_n t_n) = (\lim s_n)(\lim t_n)$.

9.5 Theorem: If $\lim s_n = s$, $(s \neq 0)$ and $s_n \neq 0$ for all n, then $\lim \frac{1}{s_n} = \frac{1}{s}$.

9.6 Theorem: If $\lim s_n = s$, $(s \neq 0)$, $s_n \neq 0$ for all n, and $\lim t_n = t$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$.

9.7 Theorem:
- (a) $\lim \frac{1}{n^p} = 0$ for $p > 0$.
- (b) $\lim a^n = 0$ if $|a| < 1$.
- (c) $\lim(n^{\frac{1}{n}}) = 1$.
- (d) $\lim(a^{\frac{1}{n}}) = 1$ for $a > 0$.

Proof. Given that $\lim a_n = a$ and $\lim b_n = b$, we can then express $\lim \frac{a_n^3+4a_n}{b_n^2+1}$ as $\frac{\lim a_n^3 + \lim 4a_n}{\lim b_n^2 + \lim 1}$. We can express the result of $\lim a_n^3$ as a^3, and $\lim b_n^2 = b^2$ by theorem 9.4. Next we can show that $\lim 4a_n = 4a$ by theorem 9.2. Then, using the result from theorem 9.3, we can show that $\lim(a_n^3 + 4a_n) = a^3 + 4a$ and $\lim(b_n^2 + 1) = b^2 + 1$. Finally, using the result from theorem 9.6, we show that:

$$\lim s_n = \frac{\lim(a_n^3 + 4a_n)}{\lim(b_n^2 + 1)} = \frac{a^3 + 4a}{b^2 + 1}$$

for $b^2 + 1 \neq 0$ for all n. □

Problem 2 *(Ross Chapter 2, 9.9(b))*

Suppose there exists N_0 such that $s_n \leq t_n$ for all $n > N_0$. Prove that if $\lim t_n = -\infty$, then $\lim s_n = -\infty$.

Discussion. Here we want to leverage the definition of divergence which is given in Ross 9.8 theorem, which states that:

> *We write $\lim s_n = -\infty$ provided for each $M < 0$ there is a number such that $n > N$ implies $s_n < M$.*

The nuance here is that we are comparing a sequence diverging to a 'negative infinity' (t_n) to another sequence that we are given to suppose is 'more negative' (s_n) after a certain point (N_0; for all $n > N_0$). In other words, we are comparing a sequence diverging to negative infinity (t_n) with another sequence (s_n) that is eventually less than or equal to it.

So if $\lim t_n = -\infty$ is true, then we know by theorem 9.8 that for every $M < 0$ there is a number N_1 such that $n > N_1$ implies $t_n < M$. We also know that there has got to be a N_0 (we suppose there is) such that $s_n \leq t_n$ for all $n > N_0$, so we can then just define an N that is greatest of the set of $\{N_0, N_1\}$ in whatever arbitrary case we take on (which we know much exist) so that we can say $s_n \leq t_n \leq M$ so that we can henceforth conclude that $s_n < M$ which is what we need to say that s_n diverges.

Proof. Let $M < 0$. We know that there exists an N_1 such that $n > N_1$ implies $t_n \leq M$.

Let $N = \max\{N_0, N_1\}$. Thus, $n > N$ implies

$$s_n \leq t_n \leq M$$

which implies

$$s_n \leq M$$

Therefore $\lim s_n = -\infty$ □

Problem 3 *(Ross Chapter 2, 9.10(b))*

Show that $\lim s_n = +\infty$ if and only if $\lim(-s_n) = -\infty$.

Discussion. We want to utilize theorem 9.8 and use a distinct M according to each case (since this is an iff statement) and prove divergence accordingly.

We will begin by show it first in the forward ($\implies$) direction: "if $\lim s_n = +\infty$, then $\lim(-s_n) = -\infty$, wherein we would let $M < 0$ so that we can say $s_n > -M$ then manipulate the expression to arrive at the conclusion that if $\lim s_n = +\infty$ then $\lim(-s_n) = -\infty$.

Then we would show the backward ($\impliedby$) direction of the iff statement such that: "if $\lim(-s_n) = -\infty$ then $\lim s_n = +\infty$", and follow the same procedure but with $M > 0$ to show the divergence toward a positive ∞ result.

Proof.
($\implies$) Let $\lim s_n = -\infty$. Let $M < 0$. For each $M < 0$ there is a number N such that $n > N$ implies

$$s_n > -M$$

which implies

$$-s_n < M$$

Thus, $\lim(-s_n) = -\infty$

($\impliedby$) Let $\lim(-s_n) = -\infty$. Let $M > 0$, For each $M > 0$ there us a number N such that $n > N$ implies

$$(-s_n) > -M$$

which implies

$$s_n > M$$

Thus, $\lim(s_n) = \infty$ □

Problem 4 *(Ross Chapter 2, 10.2)*

Prove Theorem 10.2 for bounded decreasing sequences.

Discussion. We know that Theorem 10.2 states that *"All bounded monotone sequences converge"*, and proves such by employing the definition of supS

Proof. □

Problem 5 *(Ross Chapter 2, 10.5)*

Prove Theorem 10.4(ii).

Discussion. Theorem 10.4 (ii) states: *If (s_n) is an unbounded decreasing sequence, then $\lim s_n = -\infty$.* In this exercise, you are asked to prove this statement directly, likely by adapting the argument used in part (i) of Theorem 10.4 for unbounded increasing sequences.

Proof. □

Problem 6 *(Ross Chapter 2, 10.7)*

Let S be a bounded nonempty subset of $\mathbb{R}$ such that $\sup S$ is not in S. Prove there is a sequence (s_n) of points in S such that $\lim s_n = \sup S$. See also Exercise 11.11.

Discussion. This problem asks you to construct a sequence (s_n) within S that approaches $\sup S$. Compare this to Exercise 11.11, which states a similar result but allows S to include its supremum and requires the sequence to be increasing. In contrast, this problem only assumes $\sup S \notin S$, so the challenge is to use the definition of $\sup S$ to choose terms of the sequence that get arbitrarily close to it without ever reaching it.

Proof. □